

Question 1:

The domain of the function is $(-\infty, -9) \cup (-9, 9) \cup (9, \infty)$

The range of the function is $(-\infty, -\frac{1}{81}] \cup (0, \infty)$

Detail:

To find the domain, let $t^2 - 81 \neq 0 \Leftrightarrow t \neq \{-9, 9\}$

To find the range, we have:

$$(1): t < -9 \Leftrightarrow t^2 > 81 \Leftrightarrow t^2 - 81 > 0 \Leftrightarrow \frac{1}{t^2 - 81} > 0$$

$$(2): -9 < t < 9 \Leftrightarrow 0 \leq t^2 < 81 \Leftrightarrow -81 \leq t^2 - 81 < 0 \Leftrightarrow \frac{1}{t^2 - 81} \leq -\frac{1}{81} \text{ and } \frac{1}{t^2 - 81} > 0$$

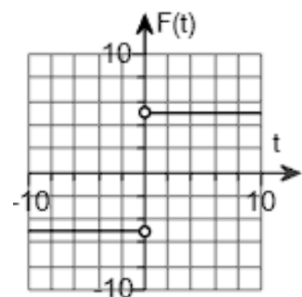
$$(3): t > 9 \Leftrightarrow t^2 > 81 \Leftrightarrow t^2 - 81 > 0 \Leftrightarrow \frac{1}{t^2 - 81} > 0$$

$$\Rightarrow \text{Range} = (1) \cup (2) \cup (3)$$

Question 2:

The natural domain of the function is $(-\infty, 0) \cup (0, \infty)$

The graph of the function:

**Detail:**

To find the domain, let $|t| \neq 0 \Leftrightarrow t \neq 0$

Draw graph:

$$\text{For } t < 0 \Rightarrow G(t) = \frac{-5t}{t} \Rightarrow G(t) = -5$$

$$\text{For } t > 0 \Rightarrow G(t) = \frac{5t}{t} \Rightarrow G(t) = 5$$

Question 3:

a.

$$f(x) = -3x \text{ if } -1 \leq x < 0$$

$$f(x) = 3 \text{ if } 0 < x \leq 1$$

$$f(x) = -\frac{3}{2}x + \frac{9}{2} \text{ if } 1 < x < 3$$

b.

$$f(x) = \frac{1}{2}x \text{ if } -2 \leq x \leq 0$$

$$f(x) = -3x + 3 \text{ if } 0 < x \leq 1$$

$$f(x) = -2 \text{ if } 1 < x \leq 4$$

Detail:

a.

Let the function: $y = ax + b$ From the graph, the first line goes from $(-1,3)$ to $(0,0)$, not touched $(0,0)$

$$\Rightarrow 3 = -a + b \text{ and } 0 = b \Rightarrow a = -3 \text{ and } b = 0 \Rightarrow y = -3x \text{ when } -1 \leq x < 0$$

The second line goes from $(0,0)$, not touched $(0,0)$, to $(1,3)$

$$\Rightarrow 3 = b \text{ and } 3 = a + b \Rightarrow a = 0 \text{ and } b = 3 \Rightarrow y = 3 \text{ when } 0 < x \leq 1$$

The third line goes from $(1,3)$ to $(3,0)$, not touched $(3,0)$

$$\Rightarrow 3 = a + b \text{ and } 0 = 3a + b \Rightarrow a = -\frac{3}{2} \text{ and } b = \frac{9}{2} \Rightarrow y = -\frac{3}{2}x + \frac{9}{2} \text{ when } 1 \leq x < 3$$

b.

Let the function: $y = ax + b$ From the graph, the first line goes from $(-2,-1)$ to $(0,0)$

$$\Rightarrow -1 = -2a + b \text{ and } 0 = b \Rightarrow a = -\frac{1}{2} \text{ and } b = 0 \Rightarrow y = -\frac{1}{2}x \text{ when } -2 \leq x \leq 0$$

The second line goes from $(0,0)$, not touched $(0,0)$, to $(1,0)$

$$\Rightarrow 3 = b \text{ and } 3 = a + b \Rightarrow a = 0 \text{ and } b = 3 \Rightarrow y = 3 \text{ when } 0 < x \leq 1$$

The third line goes from $(1,3)$ to $(3,0)$, not touched $(3,0)$

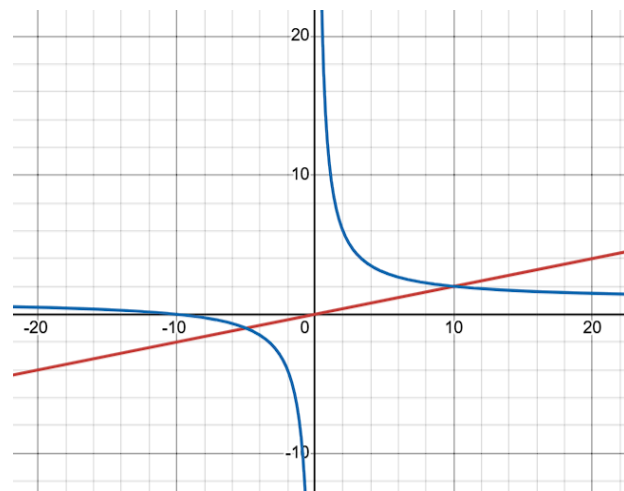
$$\Rightarrow 3 = a + b \text{ and } 0 = 3a + b \Rightarrow a = -\frac{3}{2} \text{ and } b = \frac{9}{2} \Rightarrow y = -\frac{3}{2}x + \frac{9}{2} \text{ when } 1 \leq x < 3$$

Question 4:The given function is Odd because $-h(x) = h(-x)$ **Detail:**

$$h(-x) = \frac{-9(-x)^3}{8x^2-9} = \frac{9x^3}{8x^2-9}$$

$$-h(x) = -\left(\frac{-9x^3}{8x^2-9}\right) = \frac{9x^3}{8x^2-9}$$

$$\Rightarrow -h(x) = h(-x)$$

Question 5:

The values of x for which $\frac{x}{5} > 1 + \frac{10}{x}$ is $(-5, 0) \cup (10, \infty)$

When the given inequality is rewritten so that there is a 0 on the right side and the left side is a single rational expression, the result is $\frac{x^2-5x-50}{5x}$. The numerator is 0 if $x = -5, 10$ and the denominator is 0 if $x = 0$. These values divide the real line into intervals. Using a test value from each interval confirms that $\frac{x}{5} > 1 + \frac{10}{x}$ for values of x on $(-5, 0) \cup (10, \infty)$.

Detail:

$$\frac{x}{5} > 1 + \frac{10}{x} \Leftrightarrow \frac{x}{5} - \frac{10}{x} - 1 > 0 \Leftrightarrow \frac{x^2-5x-50}{5x} > 0$$

Now we have 2 cases:

Case 1: Both $x^2 - 5x - 50 > 0$ and $5x > 0$

$$\Rightarrow x \in (0, \infty) \text{ and } x \in (-\infty, -5) \cup (10, \infty)$$

$$\Rightarrow \text{Take the intersection (using } \cap) \Rightarrow x \in (10, \infty)$$

Case 2: Both $x^2 - 5x - 50 < 0$ and $5x < 0$

$$\Rightarrow x \in (-\infty, 0) \text{ and } x \in (-5, 10)$$

$$\Rightarrow \text{Take the intersection (using } \cap) \Rightarrow x \in (-5, 0)$$

Combine Case 1 and Case 2, we have: $(1) \cup (2) = x \in (-5, 0) \cup (10, \infty)$

The numerator is 0 $\Leftrightarrow x^2 - 5x - 50 = 0 \Leftrightarrow x = -5 \text{ and } x = 10$

The denominator is 0 $\Leftrightarrow 5x = 0 \Leftrightarrow x = 0$

Question 6:

The formula for the composite function $f \circ g$ is $\sqrt{\frac{1}{x} + 6}$. The formula for the composite function $g \circ f$ is $\frac{1}{\sqrt{x+6}}$. The domain of $f \circ g$ is $(-\infty, -\frac{1}{6}] \cup (0, \infty)$.

The domain of $g \circ f$ is $(-6, \infty)$.

Detail:

$$f \circ g = \sqrt{g(x) + 6} = \sqrt{\frac{1}{x} + 6} \Rightarrow \text{Domain: } \frac{1}{x} + 6 \geq 0 \text{ and } x \neq 0$$

$$\Leftrightarrow \frac{1+6x}{x} \geq 0 \text{ and } x \neq 0$$

$$\Leftrightarrow 1 + 6x \geq 0 \text{ and } x \geq 0 \text{ and } x \neq 0$$

$$\text{or } 1 + 6x \leq 0 \text{ and } x \leq 0 \text{ and } x \neq 0$$

$$\Leftrightarrow x \geq -\frac{1}{6} \text{ and } x \geq 0 \text{ and } x \neq 0$$

$$\text{or } x \leq -\frac{1}{6} \text{ and } x \leq 0 \text{ and } x \neq 0$$

$$\Leftrightarrow x \in (0, \infty) \text{ or } x \in (-\infty, -\frac{1}{6}]$$

$$g \circ f = \frac{1}{f(x)} = \frac{1}{\sqrt{x+6}} \Rightarrow \text{Domain: } \sqrt{x+6} \neq 0 \Rightarrow x+6 > 0 \Rightarrow x \in (-6, \infty)$$

Question 7:

The domain is $(-\infty, 0) \cup (0, \infty)$

The range is $(-\infty, 0) \cup (3, \infty)$

Detail:

Domain of $f(x) \Rightarrow 1 - e^{5x} \neq 0 \Leftrightarrow e^{5x} \neq 1 \Leftrightarrow 5x \neq 0 \Leftrightarrow x \neq 0$

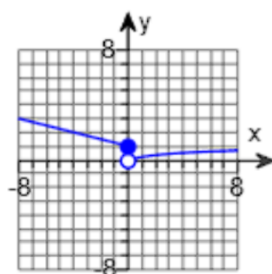
The range of $f(x)$:

(1): $x \in (-\infty, 0) \Rightarrow 0 < e^{5x} < 1 \Rightarrow 0 < 1 - e^{5x} < 1 \Rightarrow \frac{3}{1 - e^{5x}} < 0 \text{ or } \frac{3}{1 - e^{5x}} > 3$

(2): $x \in (0, \infty) \Rightarrow e^{5x} > 1 \Rightarrow 1 - e^{5x} < 0 \Rightarrow \frac{3}{1 - e^{5x}} < 0$

Combine (1) and (2) $\Rightarrow x \in (-\infty, 0) \cup (3, \infty)$

Question 8:



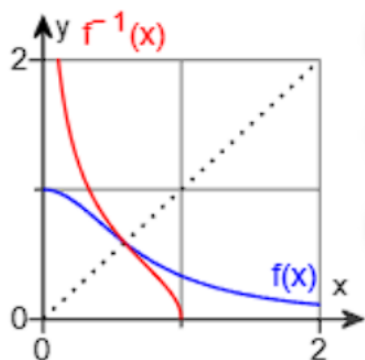
The function is 1-to-1

Detail:

Because it passes the Horizontal Line Test
(and also passes the Vertical Line Test to be a function)

Every horizontal line intersects the graph at most once

Question 9:



The domain of $f^{-1}(x)$ is $(0, 1]$

The range of $f^{-1}(x)$ is $[0, \infty)$

Detail:

$$\text{Domain of } f(x) = \frac{1}{2x^2+1}, x \geq 0$$

$$\Rightarrow 2x^2 + 1 \neq 0 \text{ and } x \geq 0 \Rightarrow x \geq 0$$

$$\Rightarrow \text{Range of } f^{-1}(x) \text{ is } x \in [0, \infty)$$

$$\text{Range of } f(x) = \frac{1}{2x^2+1}, x \geq 0$$

$$\Rightarrow 2x^2 + 1 \geq 1 \Rightarrow \frac{1}{2x^2+1} \leq 1$$

$$\text{Because } 2x^2 + 1 \text{ always } > 0 \Rightarrow \frac{1}{2x^2+1} > 0$$

$$\Rightarrow \text{Range of } f(x) = \frac{1}{2x^2+1} \text{ is } x \in (0, 1]$$

$$\Rightarrow \text{Domain of } f^{-1}(x) \text{ is } x \in (0, 1]$$

Question 10:

$$f^{-1}(x) = \frac{5x+4}{x-1}$$

$$\text{The domain of } f^{-1}(x) \text{ is } (-\infty, 1) \cup (1, \infty)$$

$$\text{The range of } f^{-1}(x) \text{ is } (-\infty, 5) \cup (5, \infty)$$

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x \text{ is True}$$

$$f(f^{-1}(x)) = \frac{\frac{5x+4}{x-1}+4}{\frac{5x+4}{x-1}-5} \text{ and } f^{-1}(f(x)) = \frac{5\frac{x+4}{x-5}+4}{\frac{x+4}{x-5}-1}$$

Detail:

$$\text{After inverse } f(x): x = \frac{y+4}{y-5}$$

$$\Leftrightarrow x(y-5) = y+4$$

$$\Leftrightarrow xy - 5x = y+4$$

$$\Leftrightarrow xy - y = 5x + 4 \text{ (all } y \text{ into one side)}$$

$$\Leftrightarrow y(x-1) = 5x+4$$

$$\Leftrightarrow y = \frac{5x+4}{x-1} \Rightarrow f^{-1}(x) = \frac{5x+4}{x-1}$$

$$\text{Domain } \Rightarrow x-1 \neq 0 \Leftrightarrow x \neq 1$$

$$\text{Range of } f^{-1}(x) = \text{Domain of } f(x) \Rightarrow x-5 \neq 0 \Leftrightarrow x \neq 5$$

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x \text{ is True}$$

$$f(f^{-1}(x)) = \frac{\frac{5x+4}{x-1}+4}{\frac{5x+4}{x-1}-5} \text{ and } f^{-1}(f(x)) = \frac{5\frac{x+4}{x-5}+4}{\frac{x+4}{x-5}-1}$$

Just replace x with $f(x)$, don't be lazy =)))